

Lecture 20 Notes

Recap and Goal: From Vector Norms to Matrix Norms

The previous lectures developed vector p -norms in $\mathbb{C}^n$ (ℓ_1 taxicab/sparsity, ℓ_2 Euclidean, ℓ_∞ Chebyshev/minimax). This lecture asks: **how do we measure how big a matrix is?** There are two fundamentally different answers.

Two Approaches to Matrix Norms

Approach 1: Flatten the Matrix into a Vector

Take the $m \times n$ matrix, stack its columns into a single mn -dimensional vector (the **vec** notation), and apply a vector norm. This approach **does not care where the entries come from** — it just stuffs them into a vector.

Using the **2-norm** of the flattened vector gives the **Frobenius norm**: the “total energy” of the matrix.

$$\|A\|_F = \left(\sum_{i,j} |a_{ij}|^2 \right)^{1/2} = \sqrt{\text{tr}(A^*A)} = \sqrt{\text{tr}(AA^*)}.$$

(Recall $\text{tr}(A^*A) = \sum_{i,k} |a_{ik}|^2$: the sum of squared magnitudes.)

Approach 2: Induced (Operator) Norms

Other norms **care about the locations** of the entries because a matrix defines a **linear mapping** $A: \mathbb{C}^n \rightarrow \mathbb{C}^m$. Picture an **input space** $\mathbb{C}^n$ and a (possibly different-dimensional) **output space** $\mathbb{C}^m$. We may measure size **differently** in each space — say a p -norm in the input and a q -norm in the output. For a nonzero input x , look at the **gain**

$$\frac{\|Ax\|_q}{\|x\|_p},$$

the factor by which the map changes the norm. Gain > 1 is amplification, < 1 is attenuation — but it depends on x (one direction might give 1.1, another 0.5). The **induced norm** is the **maximum gain** over all directions:

$$\|A\|_{q,p} = \max_{x \neq 0} \frac{\|Ax\|_q}{\|x\|_p} = \max_{\|x\|_p=1} \|Ax\|_q.$$

Scale invariance. Because $\|A(\alpha x)\|_q / \|\alpha x\|_p = \|Ax\|_q / \|x\|_p$, the ratio depends only on the **direction** of x . So we restrict the search to the **boundary of the unit p -norm ball** in the input space and ask: which boundary point produces the maximum output q -norm?

Geometric Calculation Strategy

To compute an induced norm geometrically: 1. Draw the unit $\|\cdot\|_p$ -ball boundary in the input space. 2. Map all boundary points through A to find the **image**. 3. Find the maximum $\|\cdot\|_q$ -norm over that image.

Key lemma (linear maps send line segments to line segments). Let a, b be two points with images $Aa = a'$, $Ab = b'$. Any point on the segment joining them is a **convex combination** $\lambda a + (1 - \lambda)b$ with $\lambda \in [0, 1]$, and

$$A(\lambda a + (1 - \lambda)b) = \lambda Aa + (1 - \lambda)Ab = \lambda a' + (1 - \lambda)b'.$$

So the segment from a to b maps to the segment from a' to b' . **Consequence:** for a **polytope** ball (boundaries are line segments, as for ℓ_1 and ℓ_∞), it suffices to map the **vertices** and connect them.

Induced 1-1 Norm (Worked Example)

Throughout, use the running example

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 2 \end{bmatrix}.$$

For $\|A\|_{1,1}$ we use the ℓ_1 -norm at both input and output (written just $\|A\|_1$, understood as 1-1). The unit ℓ_1 -ball is a **diamond** with vertices

$$A = (1, 0), \quad B = (0, 1), \quad C = (-1, 0), \quad D = (0, -1).$$

Map the vertices (multiplying by a standard basis vector **picks a column**):

$$A \rightarrow \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad B \rightarrow \begin{bmatrix} 2 \\ 2 \end{bmatrix}, \quad C \rightarrow \begin{bmatrix} -1 \\ 0 \end{bmatrix}, \quad D \rightarrow \begin{bmatrix} -2 \\ -2 \end{bmatrix}.$$

($B = (0, 1)$ picks column 2 = $(2, 2)$; $A = (1, 0)$ picks column 1 = $(1, 0)$.) The diamond maps to a parallelogram; the image is this **polytope** (a bounded region whose boundaries are hyperplanes; an unbounded one is a **polyhedron**, a bounded one a **polytope**). Now take ℓ_1 -norms of the image vertices:

$$\|(1, 0)\|_1 = 1, \quad \|(2, 2)\|_1 = 4, \quad \|(-1, 0)\|_1 = 1, \quad \|(-2, -2)\|_1 = 4.$$

Maximum is at B' and $-D'$: $\boxed{\|A\|_{1,1} = 4}$.

Geometric reading: if the maximum point is visually unclear, imagine expanding centered ℓ_1 -balls in the output space until the largest one still intersects the image. The last-contact points here are the images of B and D , both with ℓ_1 -norm 4.

Shortcut. This was “a heck of a thing” even for a 2×2 . It can be shown (via the Hölder inequality below) that

$$\|A\|_{1,1} = \max_j \|a_j\|_1 \quad (\text{maximum column } \ell_1\text{-norm}).$$

Here columns have ℓ_1 -norms 1 and 4; max is 4. ✓ Essential for large matrices (you cannot draw a 1000-dimensional ℓ_1 -ball).

Frobenius for comparison: $\|A\|_F = \sqrt{1 + 4 + 0 + 4} = \sqrt{9} = 3$.

Induced ∞ - ∞ Norm (Worked Example)

For $\|A\|_{\infty,\infty}$ (written $\|A\|_\infty$), the unit ℓ_∞ -ball is a **square** with vertices $(\pm 1, \pm 1)$. Map them:

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix} \rightarrow \begin{bmatrix} 3 \\ 2 \end{bmatrix}, \quad \begin{bmatrix} -1 \\ 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \begin{bmatrix} -1 \\ -1 \end{bmatrix} \rightarrow \begin{bmatrix} -3 \\ -2 \end{bmatrix}, \quad \begin{bmatrix} 1 \\ -1 \end{bmatrix} \rightarrow \begin{bmatrix} -1 \\ -2 \end{bmatrix}.$$

(($(1, 1)$ sums the columns = $(3, 2)$; $(-1, 1)$ subtracts column 1 from column 2 = $(1, 2)$.) The square maps to a parallelogram (“a parallel-parking shape”). Take ℓ_∞ -norms of the image vertices:

$$\|(3, 2)\|_\infty = 3, \quad \|(1, 2)\|_\infty = 2, \quad \|(-3, -2)\|_\infty = 3, \quad \|(-1, -2)\|_\infty = 2.$$

Maximum is $\boxed{\|A\|_{\infty,\infty} = 3}$.

Shortcut (dual to the 1-1 case). Where 1-1 used max **column** ℓ_1 -norm, ∞ - ∞ uses the maximum **row** ℓ_1 -norm:

$$\|A\|_{\infty,\infty} = \max_i \|r_i\|_1.$$

Rows have ℓ_1 -norms $\|(1, 2)\|_1 = 3$ and $\|(0, 2)\|_1 = 2$; max is 3. ✓

The bottom line: the **same matrix** gives **different norm values** for different input/output norm choices, each useful in different applications.

The Star of the Show: Induced 2-2 Norm

Using ℓ_2 at both input and output (written $\|A\|_{2,2}$ or $\|A\|_2$) gives the **maximum Euclidean gain** — “one of the most frequently used norms,” and the gateway to the **SVD**.

The Image of the Unit Sphere Is an Ellipse(oid)

The unit ℓ_2 -ball boundary in 2-D is a **circle**. Unlike the polytope cases (whose images are polytopes), the image of a sphere under a linear map is an **ellipsoid**. For invertible A , set $y = Ax$, so $x = A^{-1}y$, and the condition $\|x\|_2 = 1$ becomes $\|A^{-1}y\|_2 = 1$, i.e.

$$y^* (A^{-1})^* A^{-1} y = y^* (AA^*)^{-1} y = 1.$$

This is the equation of an **ellipse** $y^* M^{-1} y = 1$ with $M = AA^*$. The **principal semi-axes** point along the eigenvectors of M , with lengths $\sqrt{\lambda_i(AA^*)}$. The maximum ℓ_2 -norm on the ellipse is the **longest semi-axis**:

$$\|A\|_{2,2} = \sqrt{\lambda_{\max}(AA^*)} = \sigma_1(A),$$

the **largest singular value** (the nonzero eigenvalues of AA^* and A^*A coincide; either works — to be justified later).

Worked Numbers for the Running Example

For $A = \begin{bmatrix} 1 & 2 \\ 0 & 2 \end{bmatrix}$:

$$A^*A = \begin{bmatrix} 1 & 2 \\ 2 & 8 \end{bmatrix}, \quad \text{char. eqn: } \lambda^2 - 9\lambda + 4 = 0,$$

$$\lambda = \frac{9 \pm \sqrt{81 - 16}}{2} = \frac{9 \pm \sqrt{65}}{2} \approx 8.53 \text{ and } 0.47.$$

The eigenvectors give the **directions** of the principal semi-axes (e.g., the longest, v_1), and the semi-axis **lengths** are the square roots of the eigenvalues: $\sqrt{8.53} \approx 2.92$ (long axis) and $\sqrt{0.47} \approx 0.69$ (short axis). Drawing a sequence of ℓ_2 -balls of increasing radius over the ellipse, the largest one that still touches the ellipse touches it at the **ends of the longest semi-axis** — confirming that the maximum 2-norm is the longest semi-axis. Hence

$$\|A\|_{2,2} = \sqrt{8.53} \approx 2.92.$$

(The instructor’s running discussion is partly garbled around the numbers, but the eigenvalues 8.53 and 0.47 are exactly those of $A^*A = [1, 2; 2, 8]$.)

The output-side ellipse derivation in the lecture used

$$AA^* = \begin{bmatrix} 5 & 4 \\ 4 & 4 \end{bmatrix},$$

because the ellipse equation was written as $y^*(AA^*)^{-1}y = 1$. This matrix has the same characteristic equation $\lambda^2 - 9\lambda + 4 = 0$, so the same eigenvalues 8.53 and 0.47. If $AA^* = V\Lambda V^*$ and $z = V^*y$, then

$$y^*(AA^*)^{-1}y = 1 \iff \frac{|z_1|^2}{\lambda_1} + \frac{|z_2|^2}{\lambda_2} = 1.$$

So the inverse matrix contributes $1/\lambda_i$ in the ellipse equation, but the actual semi-axis lengths are $\sqrt{\lambda_i}$, not λ_i . The instructor explicitly corrected this point: the long-axis length is $\sqrt{8.53}$, not 8.53. The drawn v_1, v_2 arrows on the slide were directions of principal axes, not eigenvectors already scaled by the eigenvalues, and the picture was not to scale.

Notation Warning

$\|A\|_2$ means the **induced 2-2 norm** $= \sigma_1$, **not** the Frobenius norm. If you mean the root-sum-of-squares-of-entries, you must write $\|A\|_F$. The 2-2 induced norm is also called the **operator norm** or **spectral norm**. In some references “ $\|A\|$ ” with no subscript defaults to the operator norm; in others it defaults to Frobenius — so always check.

The Non-Invertible (Degenerate) Case — Student Question

A student asks what happens if A is **not invertible**. Then the image of the sphere is a **degenerate ellipsoid**: in 2-D with a rank-1 matrix, a line segment; in 3-D with a rank-2 matrix, an ellipse lying in the (rank-2) plane through the origin (the range space). You can reach that subspace via the **QR factorization** of A (whose Q columns give an orthonormal basis for the range) and write the ellipse in restricted coordinates. **Crucially**, the formula $\sqrt{\lambda_{\max}(A^*A)}$ still gives the maximum 2-norm **even when A is singular** — invertibility was only needed to *draw* the full ellipse, not to compute the largest semi-axis. So **invertibility is not required** for $\|A\|_{2,2} = \sigma_1$.

Shortcut Formulas Summary

Induced norm	Shortcut	Geometric meaning
$\ A\ _{1,1}$	$\max_j \ a_j\ _1$ (max column ℓ_1)	ℓ_1 -ball vertex $\rightarrow$ column
$\ A\ _{\infty,\infty}$	$\max_i \ r_i\ _1$ (max row ℓ_1)	ℓ_∞ -ball vertex $\rightarrow$ row
$\ A\ _{2,2}$	$\sigma_1 = \sqrt{\lambda_{\max}(A^*A)}$	longest semi-axis of image ellipsoid
$\ A\ _F$	$\sqrt{\sum_{ij} a_{ij} ^2}$	flatten + ℓ_2

For the running example: $\|A\|_F = 3$, $\|A\|_{1,1} = 4$, $\|A\|_{\infty,\infty} = 3$, $\|A\|_{2,2} = \sqrt{8.53} \approx 2.92$. Different numbers, different meanings — none universally “largest.”

A Few Student Questions on Defining Norms

Affine maps. A student asks whether you can define a norm for an affine map $x \mapsto Ax + b$. The instructor: you typically **lift** the affine map into a linear one by concatenation,

$$Ax + b = \begin{bmatrix} A & b \end{bmatrix} \begin{bmatrix} x \\ 1 \end{bmatrix},$$

working in the lifted space (this is how affine expressions are folded into a single matrix, used e.g. in **total least squares**). But the gain of the concatenated matrix $[A \ b]$ is **not** a true norm of the affine map, because the lifted input’s last entry is **fixed at 1** — you don’t have the freedom over that coordinate. Computing the 2-norm of $[A \ b]$ gives an **upper bound** on the affine map’s peak gain (it assumes control over that fixed coordinate), but whether the bound is achieved is unclear. The peak gain of an affine system *can* be defined, but it is a system gain, not a matrix norm.

Averaging two norms. Another student asks whether averaging norms (or “average of 2 and 4”) yields a norm. The instructor hasn’t seen it and is unsure it would satisfy the **homogeneity** (scaling) and **triangle inequality** axioms — but notes that we *do* build norms from singular values (next), so some non-obvious combinations are legitimate norms.

Preview — nuclear norm and the Netflix challenge. One especially important matrix norm to come is the **sum of the principal semi-axis lengths** (the sum of singular values / square roots of the A^*A

eigenvalues) — the **nuclear norm**. It is the key to **low-rank approximation** and powered the **Netflix challenge** (recommendation systems: recommend movies so users keep their subscription). It “burnt the brains” of many researchers in the 1990s–2000s. (The instructor jokes about reusing the same Netflix-challenge story year after year — like a friend told to “change your jokes or change your friends”; luckily the students change each term, so the jokes can repeat.) Details come with the Schatten norms in a later lecture.

Unitary-Invariant Norms

Earlier, **unitary matrices** were introduced as the matrices that **preserve the 2-norm**. Now flip the viewpoint: which **norms** are preserved under unitary maps? The 2-norm is — multiplying a vector by a unitary matrix doesn’t change its 2-norm. This is now read as a **property of the 2-norm**: it is **unitarily invariant**. By contrast, the ℓ_1 and ℓ_∞ norms are **not** unitarily invariant (a unitary rotation changes them).

Frobenius Norm Is Unitarily Invariant

For unitary U, Q :

$$\|U A Q\|_F^2 = \text{tr}((U A Q)^*(U A Q)) = \text{tr}(Q^* A^* U^* U A Q) = \text{tr}(Q^* A^* A Q).$$

Using $U^* U = I$, then cycling the trace ($\text{tr}(Q^* A^* A Q) = \text{tr}(A^* A Q Q^*) = \text{tr}(A^* A)$) and $Q Q^* = I$:

$$\|U A Q\|_F = \sqrt{\text{tr}(A^* A)} = \|A\|_F.$$

So multiplying A by unitary matrices on the left and right leaves the Frobenius norm unchanged.

2-2 (Operator) Norm Is Unitarily Invariant

The 2-2 norm is a **ratio of two 2-norms**, each unitarily invariant. Multiplying A on the right by a unitary changes the input without changing its 2-norm; multiplying on the left by a unitary changes the output without changing its 2-norm. So $\|U A Q\|_{2,2} = \|A\|_{2,2}$.

In formula form, for unitary U, Q :

$$\|U A Q\|_{2,2} = \max_{x \neq 0} \frac{\|U A Q x\|_2}{\|x\|_2} = \max_{x \neq 0} \frac{\|A Q x\|_2}{\|x\|_2}.$$

Let $z = Qx$. Since Q is unitary, $\|z\|_2 = \|x\|_2$ and z ranges over the same input space as x . Hence

$$\|U A Q\|_{2,2} = \max_{z \neq 0} \frac{\|A z\|_2}{\|z\|_2} = \|A\|_{2,2}.$$

Why this matters: both the Frobenius and 2-2 norms depend **only on the singular values** of A (which are themselves unitarily invariant). Indeed $\|A\|_{2,2} = \sigma_1$ and $\|A\|_F = \sqrt{\sigma_1^2 + \cdots + \sigma_r^2}$ (proved after SVD).

Norm Inequalities (Toward Shortcut Justifications)

Cauchy–Schwarz (Recap)

For n -dimensional vectors, the magnitude of the **Euclidean inner product** is bounded by the product of the 2-norms:

$$|\langle x, y \rangle| \leq \|x\|_2 \|y\|_2.$$

This is what lets us define the cosine of the angle (the ratio of inner product to the product of norms). The proof is short.

The Hadamard Product

Define the **Hadamard product** as **elementwise** multiplication: for same-dimension vectors (or matrices) x and y , $(x \odot y)_i = x_i y_i$. In MATLAB this is `.*` (the dot before the star); ordinary `*` is matrix multiplication. In Python (NumPy), elementwise is the default `*`, while **matrix** multiplication needs `@` (or `np.dot`) — “I like Python, but I hate that part.”

A Hölder-Type Inequality

A stronger bound than Cauchy–Schwarz uses the ℓ_1 -norm of the **Hadamard product**:

$$\|x \odot y\|_1 = \sum_i |x_i y_i| \geq |\langle x, y \rangle|,$$

because summing **absolute values** of the term-by-term products is at least the magnitude of their (signed) sum. And this is itself bounded by a product of a p -norm and a q -norm:

$$\|x \odot y\|_1 \leq \|x\|_p \|y\|_q, \quad \frac{1}{p} + \frac{1}{q} = 1.$$

The exponents are **conjugate** ($1/p + 1/q = 1$), not arbitrary. Common special cases: - $p = \infty$, $q = 1$ (the most common). - $p = 1$, $q = \infty$ (the symmetric version, used in the equalization application below). - $p = 2$, $q = 2$ (recovers Cauchy–Schwarz, since $\|x \odot y\|_1 \geq |\langle x, y \rangle|$).

So this Hölder inequality is **stronger** than bounding $|\langle x, y \rangle|$ directly.

Clarification (student Q&A). The inner product here is the **Euclidean** inner product — there is **no inner product that induces the ℓ_1 or ℓ_∞ norm** (only the 2-norm comes from an inner product). Cauchy–Schwarz in its general form holds in any **inner product space** with the **induced** norm; here we are in a normed space bounding the Euclidean inner product by **non-induced** norms (ℓ_p, ℓ_q). The full inner-product-space treatment comes later.

Use: this Hölder inequality is exactly what **justifies the shortcut formulas** for the induced 1-1 and ∞ - ∞ matrix norms (each matrix-norm entry involves a row $\times$ vector inner product, bounded by the relevant p/q norms).

For the 1-1 norm, let a_j be column j . For $\|x\|_1 = 1$,

$$\|Ax\|_1 = \sum_i \left| \sum_j a_{ij} x_j \right| \leq \sum_i \sum_j |a_{ij}| |x_j| = \sum_j |x_j| \|a_j\|_1 \leq \max_j \|a_j\|_1.$$

Equality is achieved by choosing $x = e_j$ for a column with maximum ℓ_1 -norm, so $\|A\|_{1,1} = \max_j \|a_j\|_1$.

For the ∞ - ∞ norm, let r_i be row i . For $\|x\|_\infty = 1$,

$$\|Ax\|_\infty = \max_i |\langle r_i, x \rangle| \leq \max_i \|r_i\|_1 \|x\|_\infty = \max_i \|r_i\|_1.$$

For a row attaining the maximum, choose x_j with signs matching the conjugates/signs of that row’s entries; then the bound is attained. Thus $\|A\|_{\infty,\infty} = \max_i \|r_i\|_1$.

Application: Communication-Channel Equalization

The Hölder/ ℓ_1 - ℓ_∞ inequality has a real engineering use in **channel equalization** (one of the instructor’s research areas).

Setup. In digital communications, bits are sent as a sequence of ± 1 symbols (logic 1 $\rightarrow +1$, logic 0 $\rightarrow -1$). Passing through a physical channel (reflections off “mountains,” multiple delayed copies), the signal

is **scrambled** — this is **inter-symbol interference**, modeled as **convolution** with the channel impulse response h . An ideal channel would be a **delta** (all zeros except one tap): no scrambling.

Equalizer. At the receiver we add a filter and convolve, producing an **overall channel** $g = h * (\text{equalizer})$, under our control. We want g to be as close to a **delta** as possible — i.e., **sparse**, with small ℓ_1 -norm.

The inequality at work. The equalizer output is an **inner product** of the overall channel g with the (bounded) input symbol sequence. By Hölder with $p = 1, q = \infty$: the inner product is bounded by $\|g\|_1 \|\text{symbols}\|_\infty$, and since the symbols are ± 1 (ℓ_∞ -norm = 1), **minimizing the peak (worst-case) output magnitude is equivalent to minimizing the ℓ_1 -norm of the overall channel** — which **sparsifies** g , inverting the channel without training. This connects matrix norms, the Hölder inequality, and sparsity into a practical equalization method.

Toward SVD

The image of the unit sphere under A is an **ellipsoid** whose principal semi-axes have lengths equal to the **singular values** $\sigma_1 \geq \sigma_2 \geq \dots$, point along the **left singular vectors** u_k , and correspond to input directions the **right singular vectors** v_k , with $Av_k = \sigma_k u_k$. This sphere $\rightarrow$ ellipsoid picture is the geometric heart of the **Singular Value Decomposition** — “the best thing that happened to us” — which writes a matrix as **unitary $\times$ diagonal $\times$ unitary** (two unitary matrices, one real nonneg diagonal), gives a clean picture of the four fundamental subspaces, and underlies countless algorithms and proofs. Developed in L21.

Instructor Remarks and Study Guidance

- **Frobenius $\neq$ operator norm.** $\|A\|_F = \sqrt{\sum \sigma_k^2}$ (flatten + ℓ_2); $\|A\|_2 = \sigma_1$ (induced 2-2). Both use ℓ_2 but differently; this is a classic point of confusion. Write $\|A\|_F$ explicitly.
- **Induced norm = maximum gain** over the unit input ball; restrict to the ball boundary, and for polytope balls, to the **vertices**.
- **Shortcuts:** 1-1 = max **column** ℓ_1 ; ∞ - ∞ = max **row** ℓ_1 ; 2-2 = $\sigma_1 = \sqrt{\lambda_{\max}(A^*A)}$, valid even for singular A .
- **Frobenius and 2-2 norms are unitarily invariant** (hence functions of singular values only); ℓ_1 and ℓ_∞ are not.
- **Hölder inequality** $\|x \odot y\|_1 \leq \|x\|_p \|y\|_q$ with $1/p + 1/q = 1$ is stronger than bounding $|\langle x, y \rangle|$; it justifies the matrix-norm shortcut formulas and underlies the channel-equalization sparsity application. Only the 2-norm comes from an inner product.
- **Nuclear norm** (sum of singular values) is coming — the engine of low-rank approximation and the Netflix challenge.

Source and Coverage Note

Source: `corrected/lecture20_corrected.md`.

Coverage: Two approaches to matrix norms (flatten/Frobenius vs. induced); Frobenius definition, trace formulas; induced-norm definition as maximum gain with scale-invariance and unit-ball-boundary reduction; geometric calculation strategy and the line-segment/convex-combination lemma with polytope/polyhedron terminology; worked 1-1 norm for $A = [1, 2; 0, 2]$ (diamond vertices $\rightarrow$ parallelogram, max $\ell_1 = 4$, output ℓ_1 -ball last-contact cue, max-column shortcut) and worked ∞ - ∞ norm (square $\rightarrow$ parallelogram, max $\ell_\infty = 3$, max-row shortcut); induced 2-2 norm (sphere $\rightarrow$ ellipse derivation $y^*(AA^*)^{-1}y = 1$, output-side $AA^* = [5, 4; 4, 4]$, inverse-eigenvalue coordinate equation, longest semi-axis = $\sqrt{\lambda_{\max}} = \sigma_1$, worked eigenvalues 8.53/0.47 of both AA^* and $A^*A = [1, 2; 2, 8]$, drawing/not-to-scale correction, operator/spectral-norm notation warning, non-invertible degenerate case and QR/reduced-form remark, invertibility not required); shortcut-formula summary table; student Q&As (affine-map lifting/total-least-squares upper bound, averaging norms, nuclear-norm/Netflix preview with the recurring-joke anecdote); unitary-invariant norms (2-norm

viewpoint flip, Frobenius invariance via trace proof, 2-2 invariance via explicit maximum-ratio proof, ℓ_1/ℓ_∞ not invariant); norm inequalities (Cauchy–Schwarz recap, Hadamard product with $\odot$ notation, Hölder inequality $\|x \odot y\|_1 \leq \|x\|_p \|y\|_q$ with conjugate exponents and special cases, clarification that only the 2-norm is inner-product-induced, proof sketches for the 1-1 and ∞ - ∞ shortcuts); communication-channel equalization application (ISI as convolution, overall channel sparsification, peak-minimization $\Leftrightarrow \ell_1$ -minimization via Hölder); SVD preview.